# AGML Lab: Week 3
# Logistic Regression

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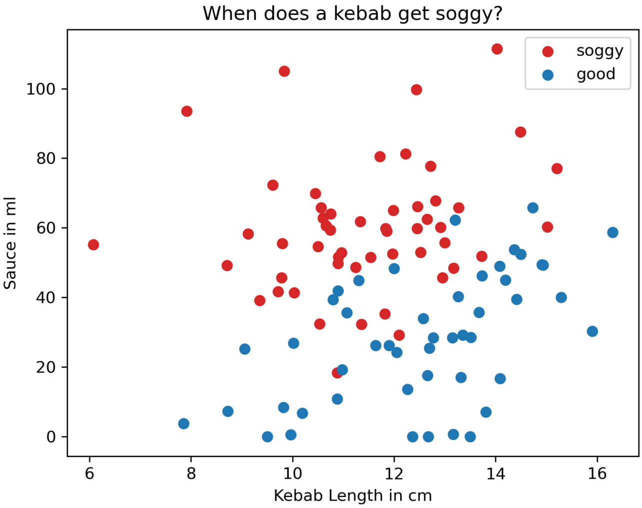
# Last Week's Assignment

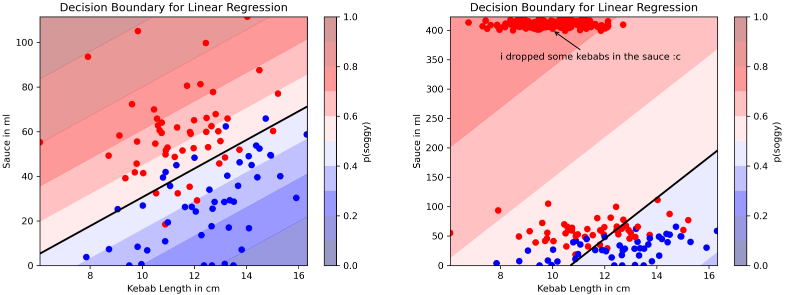


(Jupyter Notebook)

# Problem Statement

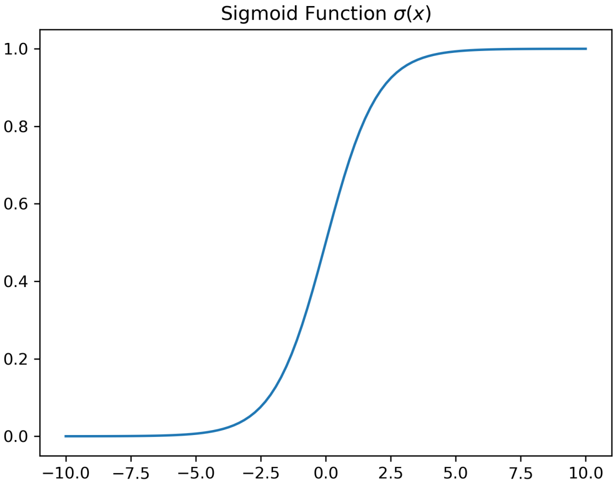


Easy! I already know how to use linear regression. I'll just use that.




**Problem:** Linear regression models the probabilities linearly.

$$p(\text{soggy}|x) = \sigma(x^\top\theta) = \frac{1}{1+e^{-x^\top\theta}}$$

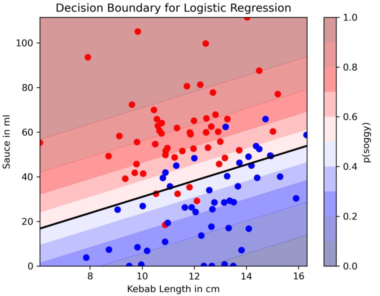


$$l(\theta) = \prod_{i=1}^{m} \underbrace{p(y_i = 1|x_i)}_{p(\text{soggy})}^{y_i} \underbrace{(1 - p(y_i = 1|x_i))}_{p(\text{not soggy})}^{1-y_i} \quad (y_i \in \{0,1\})$$

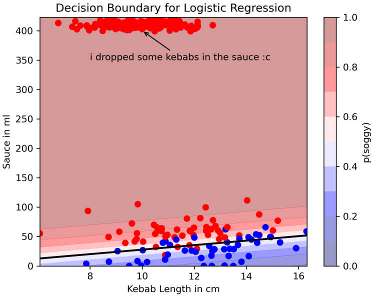
$$L(\theta) = -\frac{1}{m} \sum_{i=1}^{m} y_i \log(p(y = 1|x_i)) + (1 - y_i) \log(1 - p(y = 1|x_i))$$

There is no analytical solution anymore (but it's still convex)!
$\rightarrow$ We can use gradient descent.

# Hurray!



Decision Boundary for Logistic Regression
Sauce in ml
Kebab Length in cm
p(soggy)

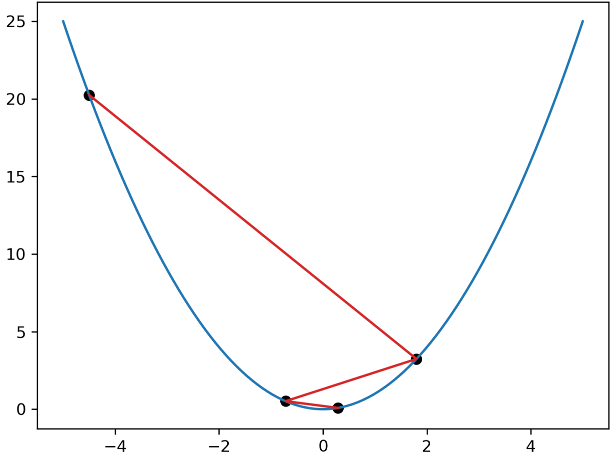



Decision Boundary for Logistic Regression
i dropped some kebabs in the sauce :c
Sauce in ml
Kebab Length in cm
p(soggy)

(Jupyter Notebook)

# Exercises

Gradient Descent: $x_{i+1} = x_i - \alpha \frac{d}{dx} f(x_i)$



$$
\begin{aligned}
L(\theta) &= -\frac{1}{m}\sum_{i=1}^{m}\underbrace{y_i\log(p(y=1|x_i)) + (1-y_i)\log(1-p(y=1|x_i))}_{K_i} \\
K_i(p) &= y_i\log(p_i) + (1-y_i)\log(1-p_i) \\
p_i(h) &= \frac{1}{1+e^{-h_i}} \\
h_i &= x_i^\top\theta \\
\frac{d\,K_i}{d\,\theta} &= \frac{d\,K_i}{d\,p_i}\frac{d\,p_i}{d\,h_i}\frac{d\,h_i}{d\,\theta}
\end{aligned}
$$